CPMS ANPR Performance Benchmarking and Accuracy Validation

Objective

Evaluate the performance and accuracy of the developed ANPR prototype against industry benchmarks and vendor-claimed performance metrics, including those published for Hikvision ANPR solutions.

Background

Commercial ANPR vendors such as Hikvision typically publish performance metrics relating to license plate detection accuracy, character recognition accuracy, vehicle capture rates, recognition speed, and performance under varying environmental conditions. The CPMS R&D; programme should independently validate these claims.

Test Scenarios

Environmental Conditions:
- Daylight
- Night-time
- Low-light environments
- Rain
- Glare and reflections
- Partial plate obstruction

Vehicle Conditions:
- Stationary vehicles
- Slow-moving vehicles
- Fast-moving vehicles
- Different vehicle types
- Dirty or damaged plates

Camera Conditions:
- High-resolution images
- Low-resolution images
- Various camera angles
- Different mounting heights
- Different zoom levels

Performance Metrics

Plate Detection Rate
OCR Accuracy
Full Plate Accuracy
False Positive Rate
False Negative Rate
Processing Time
Throughput (FPS)
CPU and Memory Utilisation

Test Dataset Requirements

Recommended Dataset:
- 500 Daytime Images
- 500 Night-time Images
- 200 Vehicle Variation Images
- 200 Difficult Scenario Images

Target Dataset Size: 1,000–2,000 labelled images

Each image should contain:
- Ground truth plate number
- Plate coordinates
- Environmental classification

Benchmark Comparison

Example Comparison:

Detection Accuracy: Internal 97.2% | Vendor 98.0%
OCR Accuracy: Internal 95.8% | Vendor 97.0%
Full Plate Recognition: Internal 93.1% | Vendor 95.0%
Processing Speed: Internal 18 FPS | Vendor 20 FPS

Vendor figures should be obtained from official documentation and validated against the specific deployment environment.

Success Criteria

- ≥95% plate detection accuracy
- ≥90% full plate recognition accuracy
- ≥15 FPS processing performance
- Acceptable day and night operation
- Performance sufficient for CPMS operational requirements

Deliverables

1. Accuracy test report
2. Performance benchmark report
3. Dataset evaluation results
4. Vendor comparison report
5. Production adoption recommendations